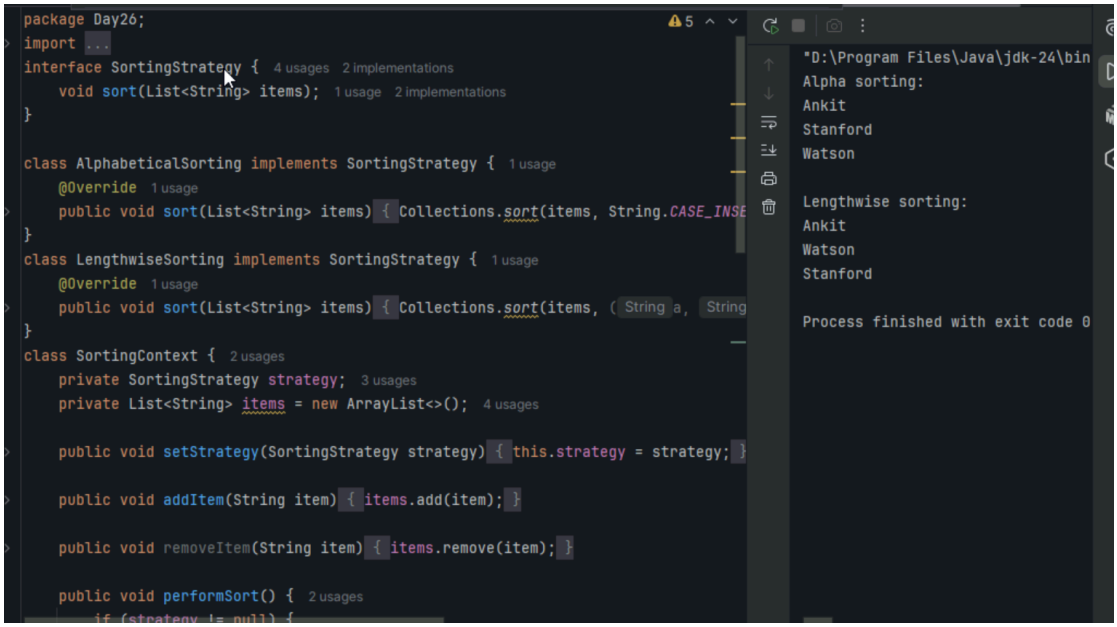
Task1



```
package Day26;
import ...
interface SortingStrategy { 4 usages 2 implementations
    void sort(List<String> items); 1 usage 2 implementations
}

class AlphabeticalSorting implements SortingStrategy { 1 usage
    @Override 1 usage
    public void sort(List<String> items) { Collections.sort(items, String.CASE_INSE
}
class LengthwiseSorting implements SortingStrategy { 1 usage
    @Override 1 usage
    public void sort(List<String> items) { Collections.sort(items, ( String a, String
}
class SortingContext { 2 usages
    private SortingStrategy strategy; 3 usages
    private List<String> items = new ArrayList<>(); 4 usages

    public void setStrategy(SortingStrategy strategy) { this.strategy = strategy; }

    public void addItem(String item) { items.add(item); }

    public void removeItem(String item) { items.remove(item); }

    public void performSort() { 2 usages
```

```
"D:\Program Files\Java\jdk-24\bin
Alpha sorting:
Ankit
Stanford
Watson

Lengthwise sorting:
Ankit
Watson
Stanford

Process finished with exit code 0
```

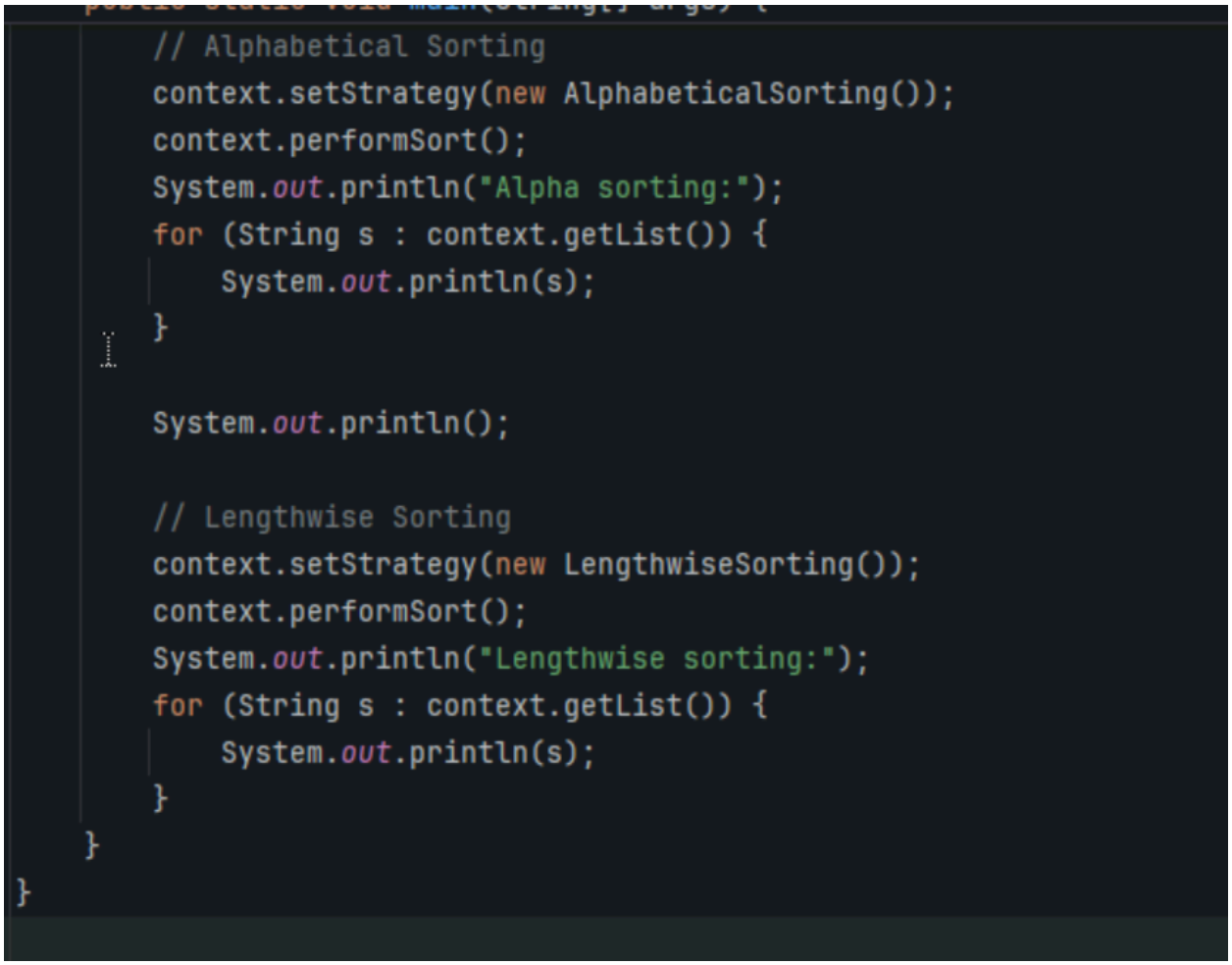
```
class SortingContext { 2 usages                                    ⚠ 5
    public void performSort() { 2 usages
            strategy.sort(items);
        } else {
            System.out.println("No sorting strategy set!");
        }
    }

    public List<String> getList() { return items; }
}

public class StrategyPatternDemo {
    public static void main(String[] args) {
        SortingContext context = new SortingContext();

        // Adding items
        context.addItem("Stanford");
        context.addItem("Ankit");
        context.addItem("Watson");

        // Alphabetical Sorting
        context.setStrategy(new AlphabeticalSorting());
        context.performSort();
        System.out.println("Alpha sorting:");
        for (String s : context.getList()) {
            System.out.println(s);
```

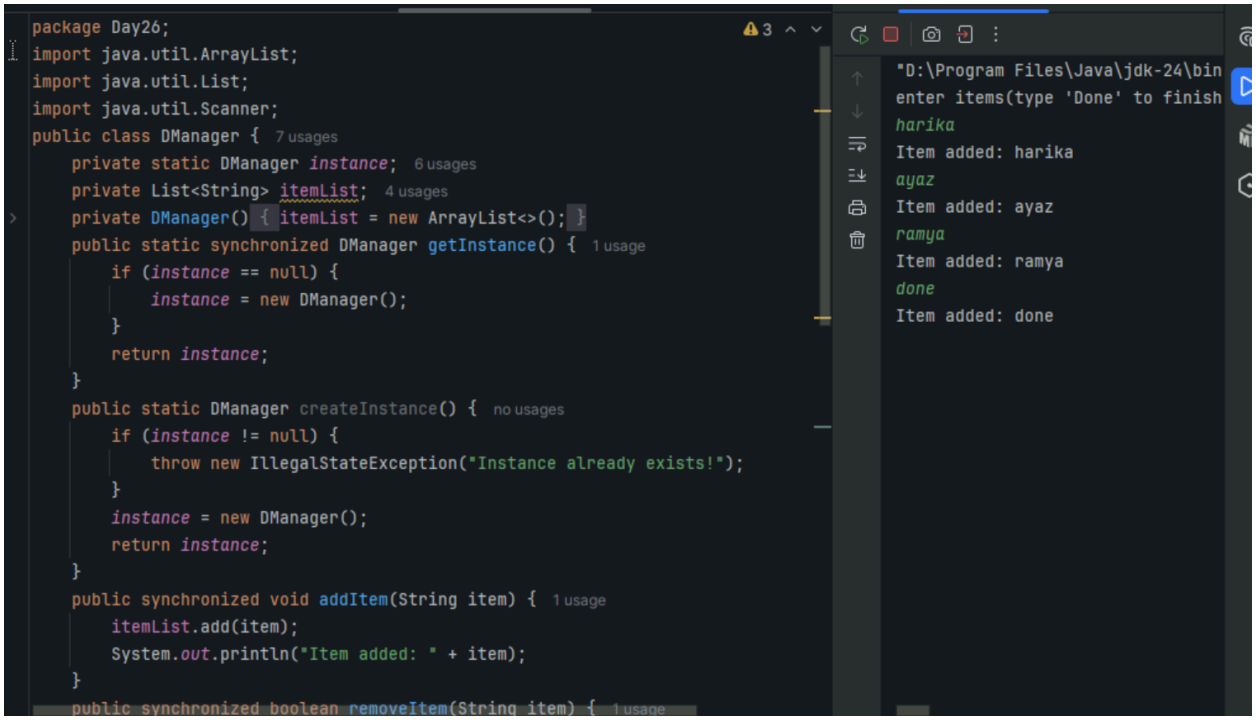


```
        // Alphabetical Sorting
        context.setStrategy(new AlphabeticalSorting());
        context.performSort();
        System.out.println("Alpha sorting:");
        for (String s : context.getList()) {
            System.out.println(s);
        }

        System.out.println();

        // Lengthwise Sorting
        context.setStrategy(new LengthwiseSorting());
        context.performSort();
        System.out.println("Lengthwise sorting:");
        for (String s : context.getList()) {
            System.out.println(s);
        }
    }
}
```

Task2



```
"D:\Program Files\Java\jdk-24\bin
enter items(type 'Done' to finish
harika
Item added: harika
ayaz
Item added: ayaz
ramya
Item added: ramya
done
Item added: done
```

```java
package Day26;
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
public class DManager {
    private static DManager instance;
    private List<String> itemList;
    private DManager() { itemList = new ArrayList<>(); }
    public static synchronized DManager getInstance() {
        if (instance == null) {
            instance = new DManager();
        }
        return instance;
    }
    public static DManager createInstance() {
        if (instance != null) {
            throw new IllegalStateException("Instance already exists!");
        }
        instance = new DManager();
        return instance;
    }
    public synchronized void addItem(String item) {
        itemList.add(item);
        System.out.println("Item added: " + item);
    }
    public synchronized boolean removeItem(String item) {
        boolean removed = itemList.remove(item);
        if (removed) {
            System.out.println("Item removed: " + item);
        }
        return removed;
    }
    public synchronized List<String> retrieveList() { return new ArrayList<>(itemList); }
}
class Singleton {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        DManager manager = DManager.getInstance();
        System.out.println("enter items(type 'Done' to finish):");
        while (true) {
            String input = scanner.nextLine();
            if (input.equals("Done")) {
                break;
            }
            manager.addItem(input);
        }
        System.out.println("item to be removed: ");
        String itemToRemove = scanner.nextLine();
        manager.removeItem(itemToRemove);
        for (String item : manager.retrieveList()) {
```

Task3

```
Class A {

        psvm() {

                Int a = 5;
                Int b = 10;
                Int c = 15:
                sout((a > b ) && (b < c))

        }
}
```

1. Compilation error
2. True
3. False
4. Runtime error

Task 4:

Finding Inheritance during requirement analysis is it important in OOAD .. why so?

1. It removes the need for encapsulation in the system design
2. It helps identify objects with the shared behavior to promote code reuse and logical hierarchy
3. It forces a flat class design improving performance by reducing polymorphic calls
4. Ensures all classes are instantiated using interfaces

Task 5:

Which characteristics best defines polymorphism in OOP?

1. Ensures each class has its own copy of data members
2. It restricts method access to specific roles within a system
3. It allows a single function or operator to behave differently based on its parameters or calling object
4. It serialized different objects into a common file format for persistence

Task 6:

Which of the following best explains the concept of data hiding in Object-Oriented Programming?

1. Data hiding means removing data from memory when no longer in use to ensure memory efficiency.

2. Data hiding involves using access specifiers to restrict direct access to class members, enabling controlled interaction through methods.

3. Data hiding refers to storing object data in secure databases during runtime.

4. Data hiding is achieved by deleting unused attributes from objects after object creation.

Task 7:

In OOAD, what is the primary value of Requirements Analysis?

1. It helps define class inheritance structure before testing

2. It identifies system behavior and user needs to model objects and interactions meaningfully

3. It configures application deployment scripts for testing

4. It automatically generates interface documentation from class files

Task 8:

Which design pattern is implemented in the following code snippet?

```
public class ClassName {

private static ClassName instance;

private ClassName() {

public static ClassName getinstance() ( if (instance = null) {

            instance = new ClassName();
           }

return instance;
}

}
```

1. Factory Method

2. Singleton

3. Prototype

4. Builder

Task 9:

Why is Interface preferred in Java when applying polymorphism over using abstract classes in many designs?

1. Interfaces enforce tight coupling between child and parent classes

2. Interfaces offer default constructors and static fields, which abstract classes cannot

3. Interfaces allow a class to inherit from multiple sources of behavior, promoting decoupling and flexibility

4. Interfaces provide direct access to private implementation logic

Task 10:

What is the role of the "Inception Phase in the Rational Unified Process?

1. It is the final phase where deployment and user training occur.

2. It defines the runtime environment for executing object oriented code

3. It helps establish the business case, scope and feasibility of the proposed systems

4. It focuses exclusively on UI design and database integration

Task 11:

What aspect of UML Diagrams makes them crucial in Object-Oriented Analysis and Design?

1. They provide detailed flowcharts for programming logic.

2. They represent runtime logs for system monitoring purposes.

3. They visually capture the structure and behavior of systems through elements like classes, objects, and interactions.

4. They replace testing frameworks by automatically generating code

Task 12:

Why is refactoring considered a continuous part of modern software development?

1. Refactoring is performed only at the end of a release cycle for documentation purposes

2. It replaces traditional debugging with automatic patching mechanisms

3. Continuous refactoring ensures that the design evolves with changing requirements, reducing technical debt and improving code health

4. Refactoring removes dependencies to minimize source control conflicts

Task 13:

OOAD, why is the Elaboration Phase important?

1. focuses on preparing production deployment pipelines

2. is where the major architectural decisions are validated through executable prototypes and risk mitigation

3. is mainly used to finalize Uf designs and wireframes

4. is dedicated to refactoring legacy code to newer patterns

Task 14:

How are Active Objects represented in object modeling

1. As static utility classes for database access

2. As objects that encapsulates encapsulate their own threat of control and asynchronously handle requests

3. As serialized containers passed between processes

4. As Java Beans used solely for UI binding

Task 15:

What makes Composite pattern useful when designing complex tree structures?

1. It replaces the use of collections to store children

2. allows treating individual objects and compositions uniformly through a common interface.

3. It automatically serializes tree objects for persistence

4. optimizes memory by removing duplicate nodes in the tree